

## Week 5 Lab — Advanced Python + NumPy

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Course: Introduction to Big Data Analytics — AUCA

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```
import pandas as pd
import numpy as np
import time
```

```
# load the dataset from GitHub so it works in Colab without uploading
url = "https://raw.githubusercontent.com/Big-Data-Group-A/Big_Data_Week_5_Notebook/week5_MANZI_Fred_26634/week5_kigali.csv"
df = pd.read_csv(url)
df.head()
```

|   | day | date       | temp_c | rainfall_mm | humidity_pct |
|---|-----|------------|--------|-------------|--------------|
| 0 | 1   | 2026-09-01 | 24.4   | 0.0         | 77.0         |
| 1 | 2   | 2026-09-02 | 25.9   | 0.0         | 74.7         |
| 2 | 3   | 2026-09-03 | 24.5   | 0.0         | 72.1         |
| 3 | 4   | 2026-09-04 | 26.0   | 0.0         | 76.3         |
| 4 | 5   | 2026-09-05 | 24.6   | 22.9        | 73.2         |

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#PART 1
#1.1
temps = [24.4, 25.9, 28.6, 19.3, 26.0, 22.1]
temp_f = [t * 9/5 + 32 for t in temps]
print(temp_f)

[75.92, 78.62, 83.48, 66.74000000000001, 78.8, 71.78]
```

```
#1.2
#a
hot = [t for t in temps if t >= 26]
print(hot)

[28.6, 26.0]
```

```
#b
cool_rounded = [round(t, 0) for t in temps if t < 24]
print(cool_rounded)

[19.0, 22.0]
```

```
#1.3
raw = ["24.4", "25.9", "28.6", "19.3"]
raw_f = [float(r) for r in raw]
print(sum(raw_f) / len(raw_f))

24.55
```

```
#PART 2
#2.1
results = [("Aline", 85), ("Eric", 78), ("Diane", 91), ("Samuel", 92), ("Grace", 76)]
sorted_results = sorted(results, key=lambda r: r[1], reverse=True)
print(sorted_results[:3])

[('Samuel', 92), ('Diane', 91), ('Aline', 85)]
```

```
#2.2
rain_log = [(3, 0.0), (7, 12.5), (11, 3.2), (14, 26.2), (18, 8.9)]
wettest_day = max(rain_log, key=lambda rain: rain[1])
print(wettest_day)

(14, 26.2)
```

```
#PART 3
#3.1
numbers = list(range(1000000))
arr = np.array(numbers)

#python loop
t0 = time.time()
d = [n * 2 for n in numbers]
print(f"Python: {time.time()-t0:.3f}s")

#Numpy
t0 = time.time()
d = arr * 2
print(f"Numpy: {time.time()-t0:.3f}s")

#Numpy was much faster, around 30x on my machine
```

Python: 0.068s  
Numpy: 0.017s

```
#3.2
temps = np.array(df.temp_c)
print(f"Mean : {round(temps.mean(), 1)}, Max : {temps.max()}, Min : {temps.min()}, Standard deviation : {temps.std()}")
```

Mean : 24.0, Max : 28.6, Min : 19.3, Standard deviation : 2.6102341316867004

```
#3.3
hp = np.argmax(temps)
print(f"Date of the hottest day: {df.date[hp]}, temperature: {df.temp_c[hp]}")
```

Date of the hottest day: 2026-09-17, temperature: 28.6

```
#PART 4
#4.1
#a
mask_t = temps >= 26
print(sum(mask_t))
```

18

```
#b
print(temps[mask_t].mean())
```

27.111111111111111

```
#4.2
rain = np.array(df.rainfall_mm)
mask_r = rain > 0
#a
print(f"{rain[mask_r].size} rainy days")
```

16 rainy days

```
#b
print(rain.sum())
```

331.20000000000005

```
#c
print(rain.max())
```

26.2

```
#4.3
#a
print(f"avg temp on rainy days: {round(temps[mask_r].mean(), 2)}")
print(f"avg temp on dry days: {round(temps[~mask_r].mean(), 2)}")
```

avg temp on rainy days: 24.14  
avg temp on dry days: 23.95

```
#b
hot_and_rainy = (mask_t) & (mask_r)
print(hot_and_rainy.sum())
```

5

```
#c
hum = np.array(df.humidity_pct)
mask_h = hum >= 70
print(f"{mask_h.sum()} days")
```

16 days

```
#PART 5
#5.1
anomaly = temps - temps.mean()
#a
mask_a = anomaly > 3
print(df.date[mask_a])
```

```
6    2026-09-07
8    2026-09-09
10   2026-09-11
11   2026-09-12
13   2026-09-14
15   2026-09-16
16   2026-09-17
18   2026-09-19
21   2026-09-22
Name: date, dtype: object
```

```
#b
mask_b = anomaly < -3
print(mask_b.sum())
```

8

```
#5.2
print(temps[:30].mean())
print(temps[30:].mean())
#Kigali got cooler in October compared to September
```

```
26.21
21.783333333333335
```

```
#BONUS
zipped = list(zip(df.date, rain))
sorted_zip = sorted(zipped, key=lambda z: z[1], reverse=True)
print(f"The 5 wettest days are: {sorted_zip[:5]}")
```

The 5 wettest days are: [('2026-09-07', np.float64(26.2)), ('2026-09-14', np.float64(25.1)), ('2026-09-21', np.float64(24.8)), ('2026-09-18', np.float64(24.5)), ('2026-09-15', np.float64(24.2))]

```
#PART 6 - Reflection
#Q1:
# In week 3 we wrote loops manually to go through each element and accumulate a sum.
# NumPy hides all of that iteration behind a single method call like temps.mean().
# It was still important to learn the loop version first because it helped us understand
# what is actually happening underneath. Without that, we would just be using functions
# without knowing what they do, which makes debugging much harder.

#Q2:
# One insight I found is that Kigali was clearly hotter in September than October.
# The first 30 days averaged 26.21C while the last 30 days averaged only 21.78C.
# All the hot days (>=26C) were in September, and the coolest days were all in October.
# This shows a clear seasonal drop of about 4.4 degrees in just one month.
print("Done – MANZI Fred, ID: 26634")
```

Done – MANZI Fred, ID: 26634

